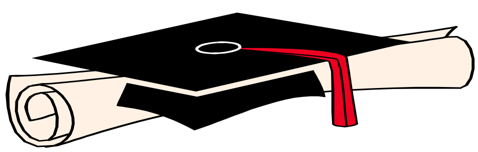
EPC

Elementary Programming with C

Semester: 1	**The Exam Code:** 16
Duration: 90 minutes	**Total Marks:** 20



Practical Examination Paper

Do not write on this question paper and return it to the Invigilator after the examination.

C Programming Assessment: Cosmetics Inventory Management System with Pointer Concepts

Total Points: 20

Instructions:

- Ensure your code is well-structured, easy to read, and includes appropriate comments explaining your logic.
- Use pointers where applicable to demonstrate understanding of memory management in C.
- Handle errors appropriately where necessary.
- Follow C conventions for naming and coding practices.

Question 1: Basic Arithmetic Operations with Error Handling [3 Points]

a) Cosmetic Product Price Calculator [3 Points]

Task:
Write a C program that prompts the user to enter two prices of cosmetic products and an operator (+, -, *, /). The program should then perform the corresponding arithmetic operation and display the result. Include error handling for division by zero.
Example:
Input:

```
Enter first price: 15.99
Enter second price: 3.5
Enter operator (+, -, *, /): /
```

Output:

```
Result: 4.57
```

Points Distribution:

- Correct arithmetic operation based on the operator: **2 Points**
- Correct error handling for division by zero: **1 Point**

Question 2: Nested Loops and Conditional Statements [3 Points]

a) Discount Table for Cosmetic Products [3 Points]

Task:
Write a C program that prints a discount table for a cosmetic product. The table should display the price of the product after applying discounts from 5% to 50% in increments of 5%, for quantities from 1 to 10.
Example Output:

```
     5%     10%     15%     20%    ... 50%
1      $9.5   $9.0    $8.5    ...
2      $19    $18     $17    ...
...
10     $95    $90     $85    ...
```

Points Distribution:

- Correct use of nested loops: **2 Points**
- Correct formatting of output: **1 Point**

Question 3: Arrays, Pointers, and Searching [4 Points]

a) Linear Search in Cosmetic Product Inventory with Pointers [4 Points]

Task:
Write a C program that allows the user to input 10 product prices into an array. Use a pointer to traverse the array and perform a linear search for a specific price entered by the user.
If the price is found, print the index at which it was found; otherwise, print a message indicating that the price was not found.
Example:
Input:

```
Enter 10 product prices: 15.5 20.0 25.5 10.0 5.0 18.75 22.5 30.0 12.0
8.5
Enter a price to search: 18.75
```

Output:

```
Price found at index: 5
```

Points Distribution:

- Correct use of pointers for array traversal: **1 Point**
- Correct implementation of linear search: **2 Points**
- Correct output based on search result: **1 Point**

Question 4: Functions and Recursion [4 Points]

a) Recursive Discount Calculation [4 Points]

Task: Write a recursive function `void applyDiscount(double *price, int percentage)` that calculates the discounted price for a cosmetic product.

The function should reduce the price by a given percentage using pointers to update the price directly.

Use this function in a program that takes a product price and discount percentage as input from the user and prints the discounted price.

```
Enter price: 100
Enter discount percentage: 20
```

Output:

```
Discounted Price: 80
```

Points Distribution:

- Correct implementation of the recursive function: **3 Points**
- Correct use of the function in the main program: **1 Point**

Question 5: String Manipulation and Pointers [3 Points]

a) Reverse a Product Name Using Pointers [3 Points]

Task:
Write a C program that takes a product name as input and prints the product name in reverse order. Use pointers to traverse the string and reverse it. Do not use any built-in string functions for reversing the string.
Example:
Input:

```
Enter product name: Lipstick
```

Output:

```
Reversed product name: kcitspiL
```

Points Distribution:

- Correct input handling: **1 Point**
- Correct logic to reverse the string using pointers: **2 Points**

Question 6: Structures and Sorting [3 Points]

a) Sort Cosmetic Products by Price Using Structures [3 Points]

Task:
Define a structure `CosmeticProduct` with fields:

- `name` (string)
- `id` (int)
- `price` (float)

Write a C program that allows the user to input data for 3 cosmetic products, stores them in an array of `CosmeticProduct` structures, and then sorts and displays them by price in ascending order.
Example:
Input:

```
Enter name, id, and price for product 1: Lipstick 101 15.99
Enter name, id, and price for product 2: Foundation 102 25.5
Enter name, id, and price for product 3: Mascara 103 12.75
```

Output:

```
Sorted products by price:
Product: Mascara, ID: 103, Price: $12.75
Product: Lipstick, ID: 101, Price: $15.99
Product: Foundation, ID: 102, Price: $25.5
```

Points Distribution:

- Correct structure definition: **1 Point**
- Correct sorting and display logic: **2 Points**

This assessment introduces essential C programming concepts such as pointers, recursion, arrays, and structures, providing a comprehensive test of your coding skills.